

Predicting Economic Damage of Natural Disasters

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Invalid Date

Data processing

Introduction

Background:

As global temperatures rise, climate-related disasters have increased in both frequency and severity over the past several decades ([World Wild Life](#)). Extreme weather events such as floods, droughts, hurricanes, etc. impose a huge human and real economic costs across the world. It is imperative that we understand the determinants of disaster severity (in terms of economic damages) for better policy planning and mitigative strategies.

The Emergency Events Database (EM-DAT), maintained by the Centre for Research on the Epidemiology of Disasters (CRED), keeps an internationally standardized data on natural disasters worldwide. The information includes disaster type, location, affected population, deaths, injuries, and economic damages.

Research Question:

How do different disaster characteristics (such as geographic location, time of year, type, and total casualties incurred) predict the total economic damage caused by natural disasters in USD?

- Specifically, we examine whether disaster type, location, timing, duration, and measures of human impact (deaths, injuries, and affected population) are associated with higher reported economic losses.

Motivation:

We were first drawn to this research question because we were intrigued by this data; we were amazed by the granularity and the surprising breadth of information in this data. However, we realized that doing research on this matter could have real, practical uses.

First, identifying predictors of large economic damage is critical for both public and private institutions. Governments and international organizations allocate substantial resources towards mitigation and recovery efforts, yet the determinants of high-cost disasters vary across event types and regions. Same applies for insurance companies and hedge fund traders that try modeling key determinants relevant to a disaster for their comparative advantage against peers in the market.

By analyzing global disaster data, this project aims to quantify which observable characteristics are most strongly associated with economic loss. We hope that these results may provide insight into which types of disasters and impact measures are most economically destructive.

Hypothesis Reasoning:

Based on economic reasoning and relevant research, we expect that...

1. Disasters with higher number of affected individuals (absolute number, number of deaths and injuries) will have a positive relationship with economic damage.
2. Certain disaster types will incur systematically higher economic damage than that of others.
3. The level of damage dollars will differ based on countries because of their respective infrastructure that exists to mitigate damage.

Data Description:

This data was obtained from the Emergency Events Database, otherwise known as [EM-DAT](#), which is maintained by the Centre for Research on the Epidemiology of Disasters (CRED). The goal of the database is to compile standardized information on natural disasters that occur around the world. According to the information on metadata, an event is included in EM-DAT if one or more of the following criteria is met:

- 10+ people killed
- 100+ people affected
- Declaration of a state of emergency
- Solicitation of international assistance

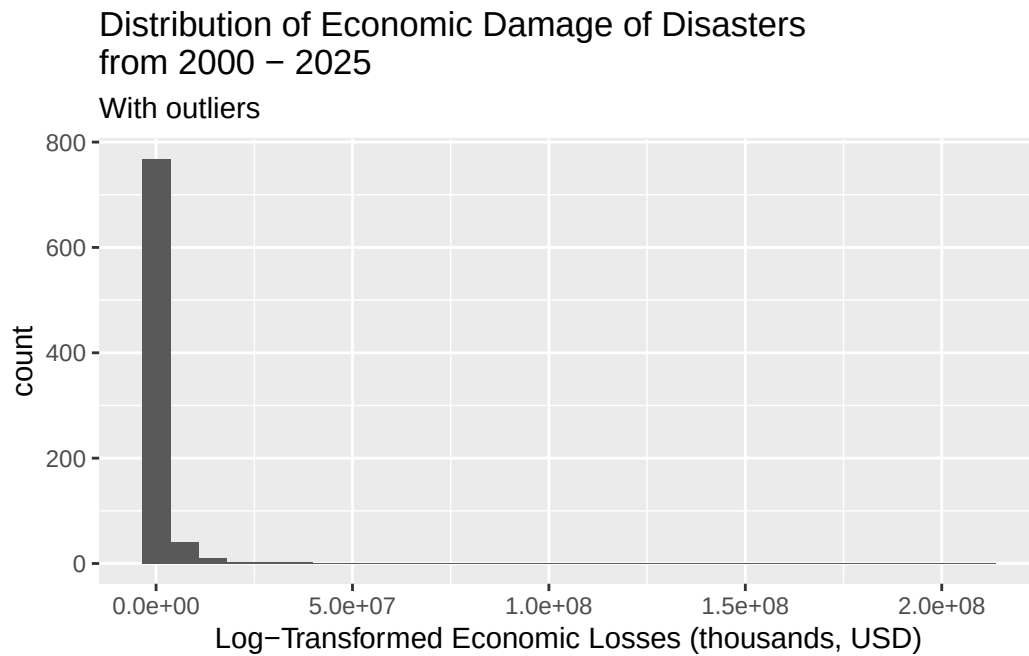
The observations describe different disasters that occurred around the world. Of each disaster, data on the number of people injured, affected, homeless, or killed, the total economic damages, type of disaster, exact location, start date and end date of the disaster were collected.

The data dictionary can be found [here](#).

Limitations and Concerns

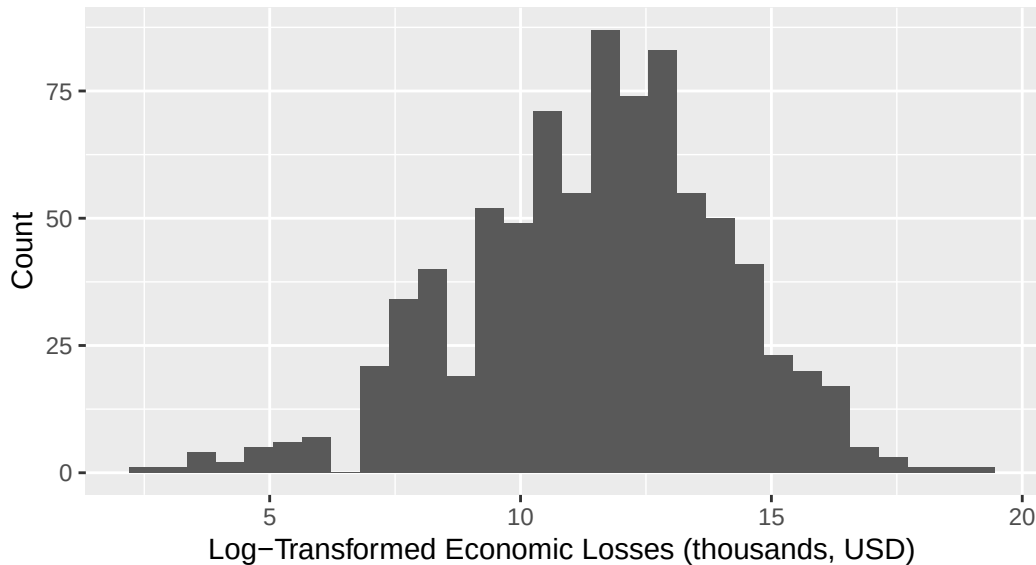
There are a considerable number of outliers in our data set with respect to total economic damage, duration in days, and many other variables. This motivated us to apply the logarithmic function to those variables. These outliers, however, should still be considered in the larger-picture analysis.

Exploratory Data Analysis



Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
12	20000	109576	1519107	559750	210000000

Distribution of Economic Damage of Natural Disasters from 2000 – 2025



Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
2.485	9.903	11.604	11.493	13.235	19.163

[1] 2.606016

The distribution of the unchanged response variable is right-skewed and unimodal. The center is best represented by the median, which is \$109,576,000, and the spread is best represented with the IQR, which is \$539,750,000 (559,750,000 - 20,000,000). There are 123 (828-705) outliers present in the upper end of our distribution.

The distribution of the logged response variable is unimodal and symmetric. The center is best represented by the mean, which is \$98,027,173.52 ($e^{11.493} * 1000$), and the spread is best represented with the standard deviation, which is \$13544.98 ($e^{2.606016} * 1000$). There are outliers present in our distribution. **We will be working with the logged response variable**, log of total economic damage in thousands of USD, because without the log, the outliers skew the data heavily.

Predictors of interests:

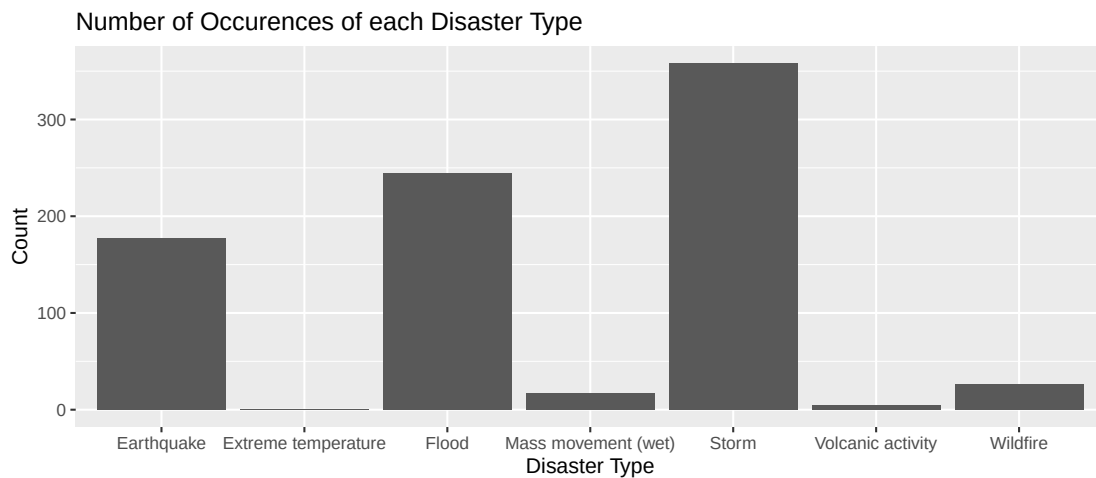
1. disaster_type - the type of disaster (categorical)

```
# A tibble: 7 x 1
  disaster_type
```

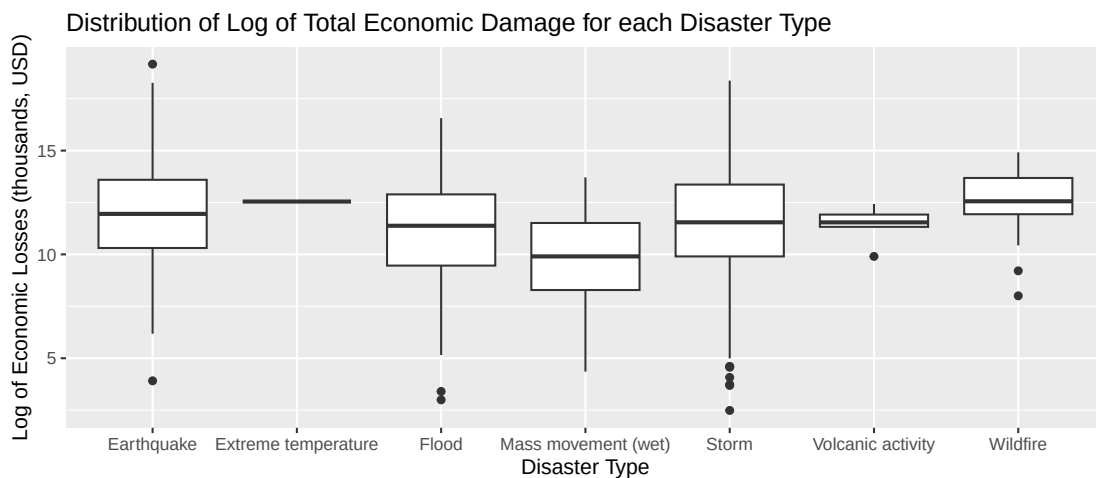
```

<chr>
1 Flood
2 Storm
3 Earthquake
4 Mass movement (wet)
5 Wildfire
6 Volcanic activity
7 Extreme temperature

```



Storms, floods, and earthquakes are the most common disaster types, composing most of the data. Storms are the most common. Extreme temperature, wet mass movements, volcanic activity, and wildfires are the least common disaster types, with extreme temperature being the least common.

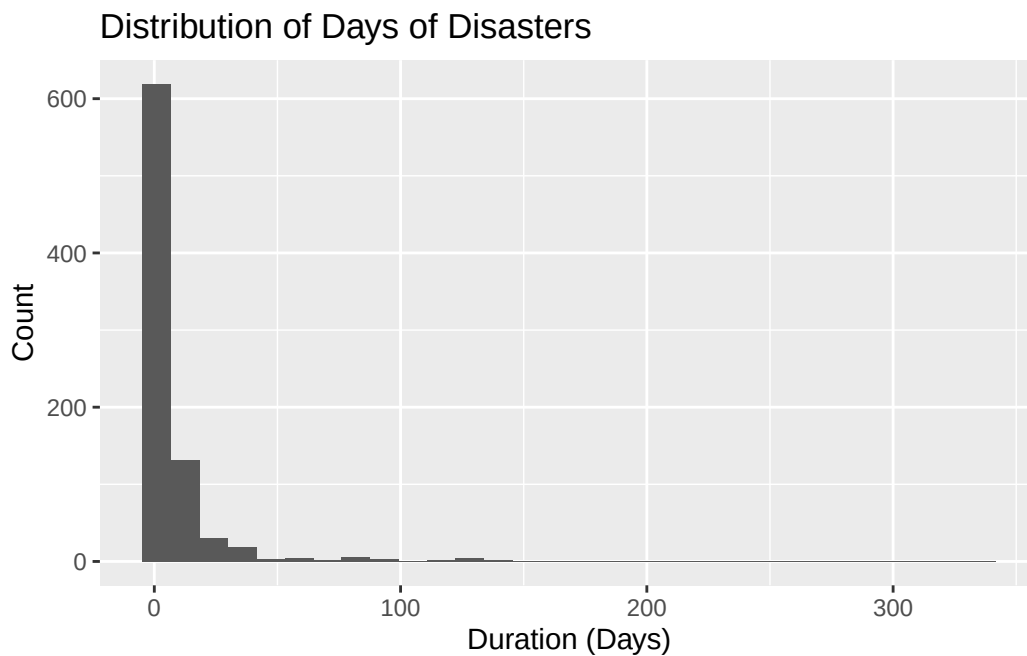


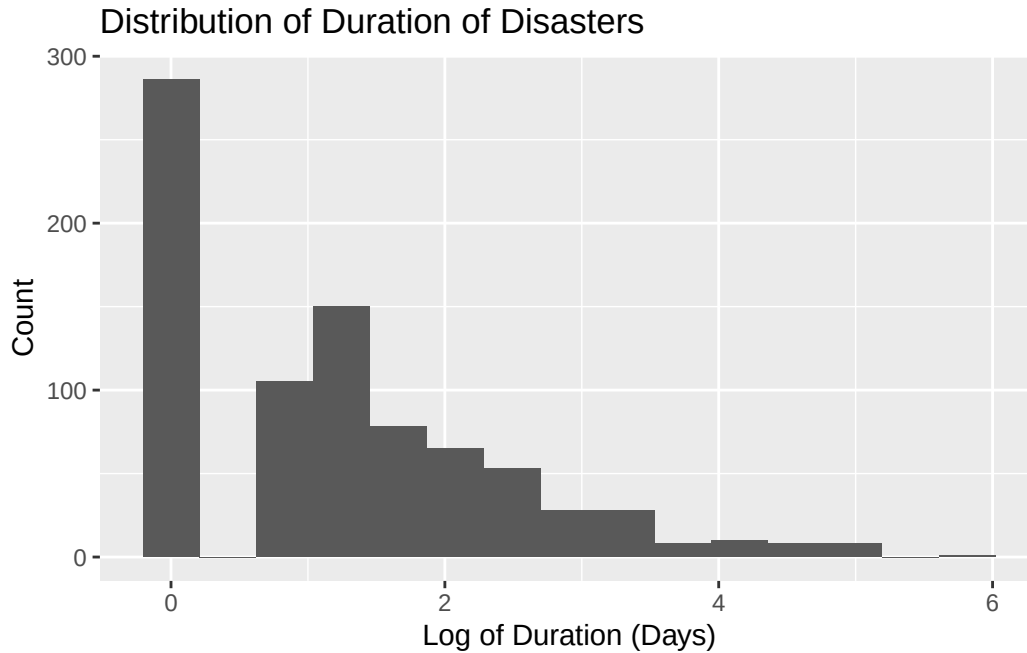
```
# A tibble: 7 x 3
```

	disaster_type	median	IQR
	<chr>	<dbl>	<dbl>
1	Earthquake	12.0	3.28
2	Extreme temperature	12.5	0
3	Flood	11.4	3.43
4	Mass movement (wet)	9.90	3.23
5	Storm	11.5	3.46
6	Volcanic activity	11.5	0.592
7	Wildfire	12.6	1.75

Wildfires have the highest median log of total economic damage (in thousands of dollars), which is 12.558857, and thus, normal total economic damage, which would be 284604.84 thousands of dollars ($e^{12.558857}$). The median of earthquakes is slightly lower, and then the median of volcanic activities, floods, and storms, which all have similar values, slightly below the median of earthquakes. Wet mass movements have the lowest median (log of) total economic damage. Storms, floods, and earthquakes have the highest variability in the (log of) total economic damage.

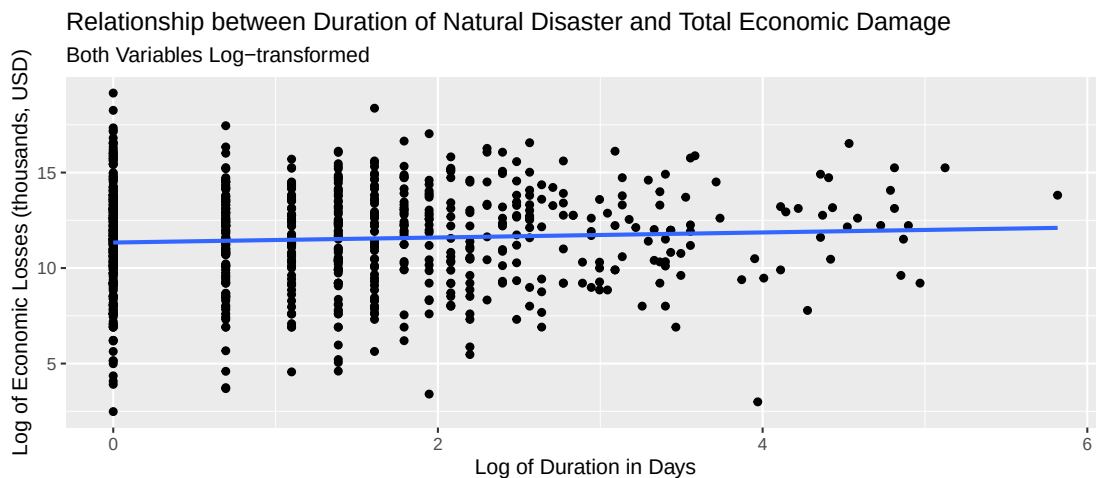
2. duration - duration in days of the disaster (quantitative)





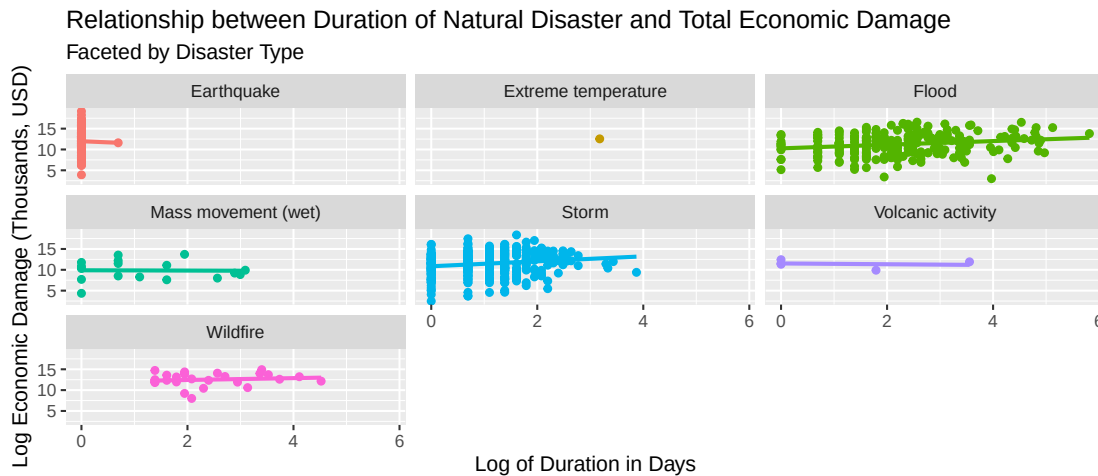
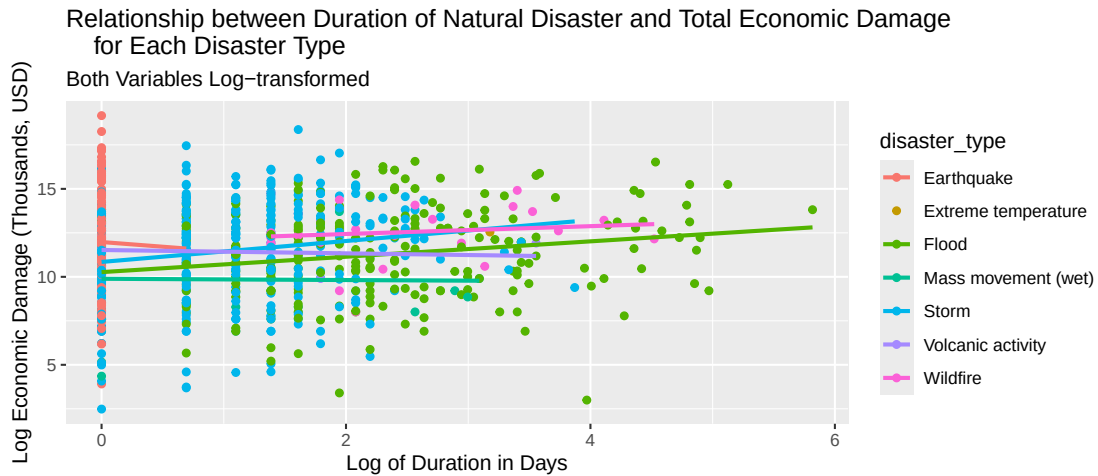
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000	0.000	1.099	1.187	1.946	5.817

Because of the extreme outliers and extreme skew in the histogram of the duration (in days) of each disaster, we decided to explore the log of the duration of the disaster. The histogram of the log of the duration has right skew and is bimodal. This makes sense because a lot of the disasters only last 1 day long due to their inherent nature, such as earthquakes. The median is 1.099 log duration in days, and the IQR is 1.946 log duration in days. There are outliers in the upper end of our distribution.



```
# A tibble: 1 x 1
  r_squared
    <dbl>
1  0.00360
```

There seems to be a weak positive linear relationship between the log of duration in days with log of total economic damage (in thousands of dollars). The r-squared value is 0.003596655. There is a lot of variability in the response variable when the log of duration in days is low, but the variability decreases as the x value increases. There are outliers present throughout the graph.

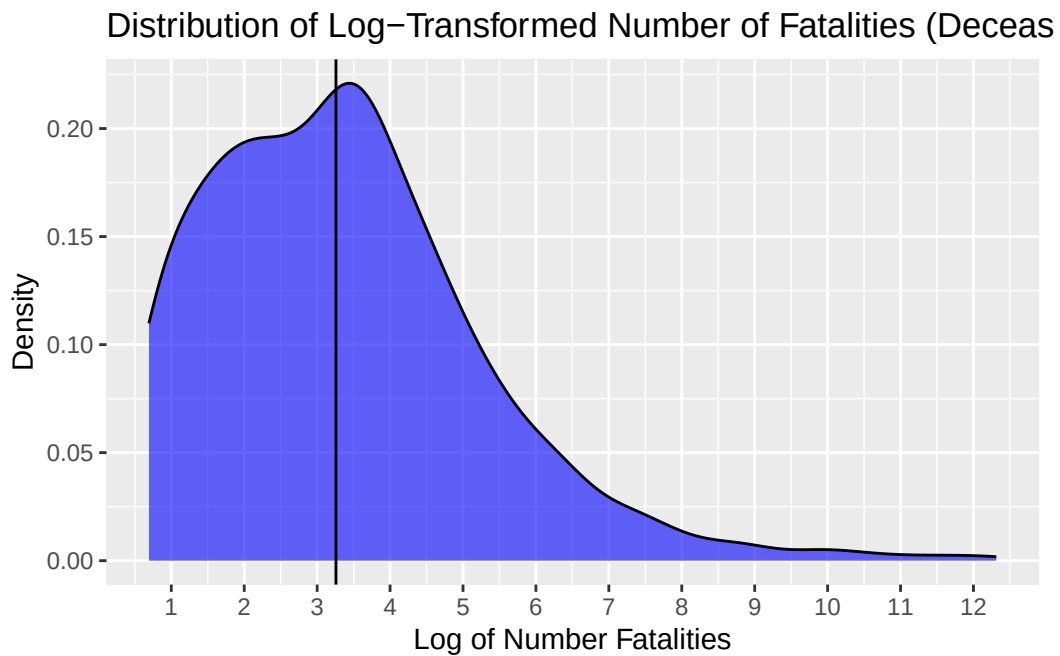


After visualizing the potential interaction effect between log duration of days and type of disaster, it is evident that an increase in the log duration in days has a stronger effect for storms and floods, compared to wildfires, volcanic activities, and wet mass movements,

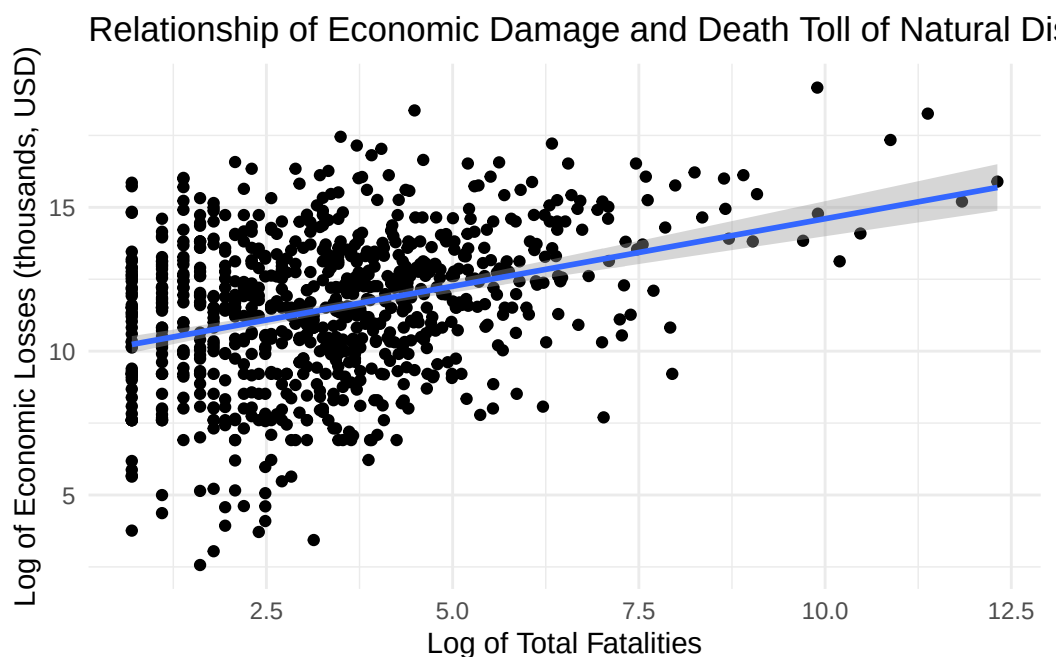
which is shown by the slightly steeper positive slope. This makes logical sense as well. We will consider putting this interaction effect into our model because of this visualization.

3. total_deaths - Total fatalities (deceased and missing combined) (quantitative)

Statistic	Value
Min.	1.000
1st Qu.	6.000
Median	25.000
Mean	921.675
3rd Qu.	80.000
Max.	222570.000



Because of the extreme skewness and outliers in the data of total fatalities (of each natural disaster), we decided to explore a log transformation of the data. In the density plot above, we can see that the overall distribution is unimodal and has a distinct right skew. From the summary statistics, we can tell that while the median number of fatalities is 25 people, some grand disasters have a catastrophic number of deaths, with the highest number being 222570 people.



[1] 0.1159678

There is a clear positive relationship between total deaths and economic damage. Disasters with higher fatalities generally correspond to higher economic losses (USD). However, the relationship is noisy and dispersed, indicating that deaths alone may not be a strong predictor of damage and that further analysis needs to be done.

Differences across disaster types are visible. Earthquakes and storms tend to generate higher economic damage at comparable fatality levels, while floods and mass movements appear more concentrated at lower damage levels. This suggests that the impact of fatalities on economic damage likely varies by disaster type. This indicates potential interaction effects are present.

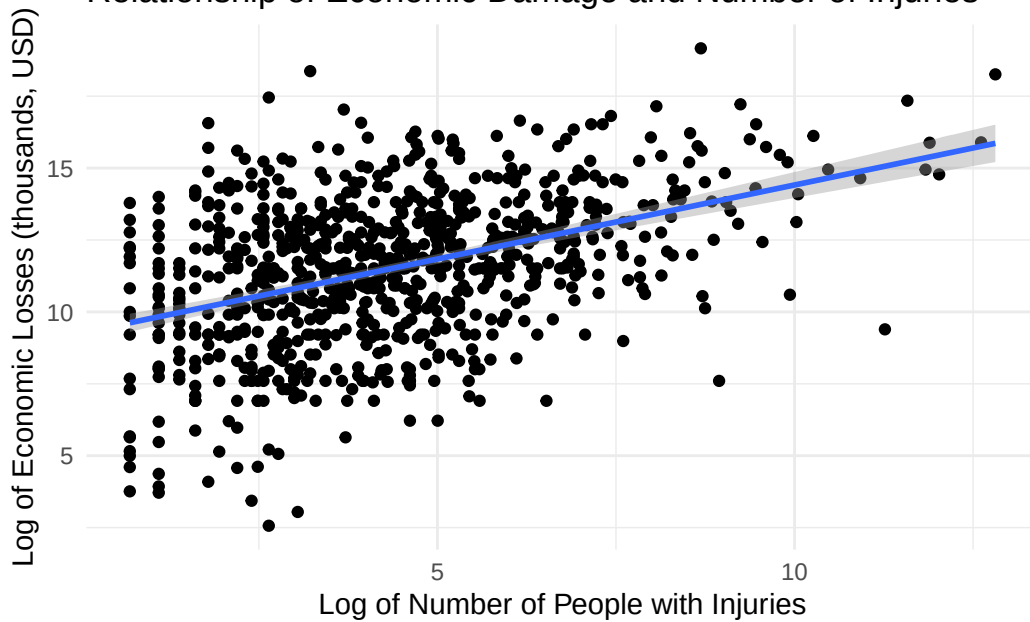
4. no_injured - Number of people with physical injuries, trauma, or illness requiring immediate medical assistance due to the disaster (quantitative)

Statistic	Value
Min.	1.00
1st Qu.	14.00
Median	60.00
Mean	2338.36
3rd Qu.	266.25
Max.	366596.00

Distribution of Log-Transformed Number of People With Injuries



Relationship of Economic Damage and Number of Injuries



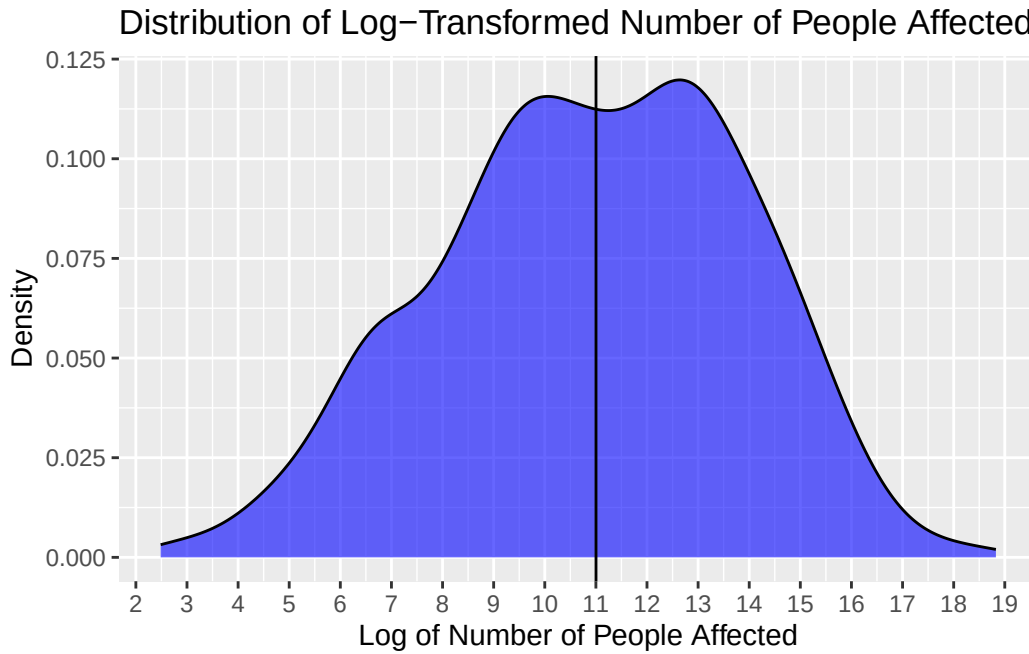
[1] 0.1828717

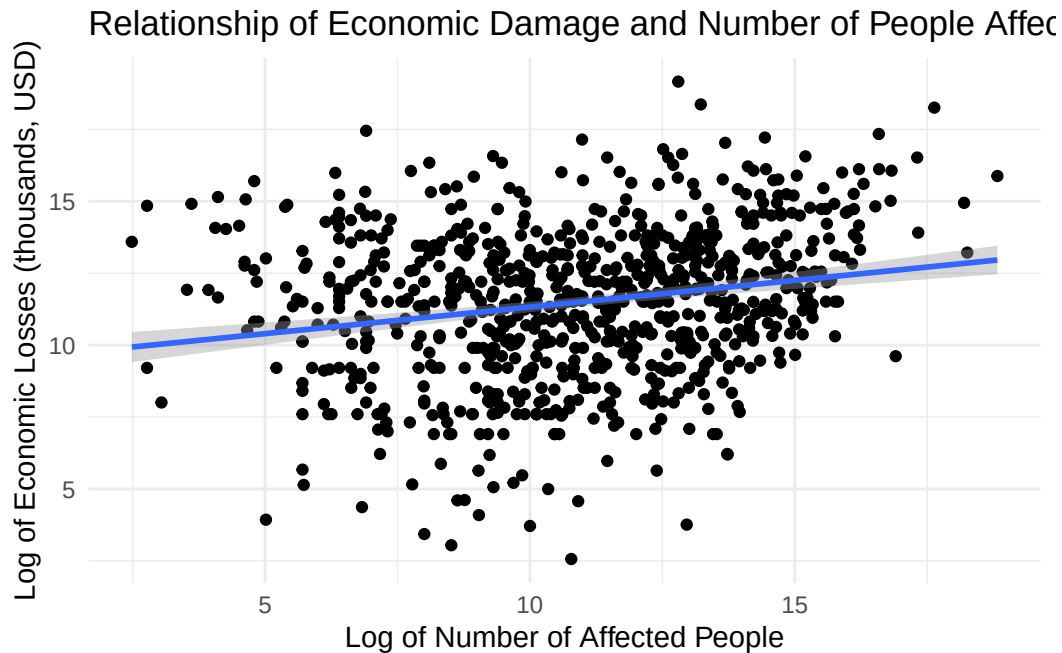
R-squared = 0.18

The story is similar to that of the previous predictor, `total_deaths`. There is a clear unimodal shape with a right skew, and we know that this skewness has been alleviated by the log transformation (so in reality, much larger). It makes sense that the median is higher at 60 people because, common-sensically, there will always be more people injured than killed at any given natural disaster. This corresponds to the maximum number being higher than that of `total_deaths` at 366596 people.

5. `no_affected` - Number of people requiring immediate assistance due to the disaster (quantitative)

Statistic	Value
Min.	11
1st Qu.	7000
Median	60000
Mean	1325411
3rd Qu.	500250
Max.	150000000





[1] 0.04448019

R-squared = 0.044

We again applied a log transformation to better visualize the distribution. In the density plot above, the distribution of the log-transformed values appears roughly unimodal, though there is still a right skew, considering the distribution before the log transformation. Compared to fatalities, the distribution of affected individuals is more spread out, reflecting the wide range of disaster scales – from local events to those impacting entire countries.

From the summary statistics, there is a gap between the median (60,000 people) and the mean (over 1.3 million people), indicating a big right tail driven by a small number of extremely large disasters. The maximum value of 150,000,000 people affected highlights the scale at which certain disasters can disrupt entire populations. This makes sense considering the nature of natural disasters: while many events are relatively small, large-scale disasters can impact vast populations through displacement and infrastructure damage.

Methodology

Splitting the data -

Determining if multicollinearity is present -

To determine if there is multicollinearity among the variables, we calculated the variance inflation factor (VIF) for each predictor variable.

log(no_affected)	log_duration_days
1.599947	1.994663
disaster_typeExtreme temperature	disaster_typeFlood
1.044478	3.561442
disaster_typeMass movement (wet)	disaster_typeStorm
1.252575	2.464263
disaster_typeVolcanic activity	disaster_typeWildfire
1.039201	1.432605
log(total_deaths)	log(no_injured)
2.204314	2.069495

Because all of the VIF values are below 5, we concluded that multicollinearity is not prevalent among the variables, and we opted to include all of them in the model.

Determining the best model -

We were curious to see if a model with an interaction effect between log_duration_days and disaster_type would be a better model, so we conducted cross validation to see which model performed better.

```
# A tibble: 2 x 6
  .metric .estimator mean      n std_err .config
  <chr>   <chr>      <dbl> <int>   <dbl> <chr>
1 rmse    standard    2.23     8  0.0549 pre0_mod0_post0
2 rsq     standard    0.229     8  0.0230 pre0_mod0_post0
```

```
# A tibble: 2 x 6
  .metric .estimator mean      n std_err .config
  <chr>   <chr>      <dbl> <int>   <dbl> <chr>
1 rmse    standard    2.23     8  0.0566 pre0_mod0_post0
2 rsq     standard    0.230     8  0.0245 pre0_mod0_post0
```

At this point, we are going to move forward with the model with log no_affected, log duration_days, disaster_type, log total_deaths, and log no_injured without any interaction effects because after conducting cross-validation, we discovered that this model has a higher r-squared compared to the model with the interaction effect.

term	estimate	std.error	statistic	p.value
(Intercept)	8.898	0.440	20.239	0.000
log(no_affected)	0.007	0.039	0.185	0.853
log_duration_days	0.285	0.107	2.667	0.008
disaster_typeExtreme temperature	-0.072	2.274	-0.032	0.975
disaster_typeFlood	-0.593	0.373	-1.590	0.112
disaster_typeMass movement (wet)	-1.089	0.673	-1.619	0.106
disaster_typeStorm	0.000	0.282	0.001	0.999
disaster_typeVolcanic activity	0.107	1.311	0.081	0.935
disaster_typeWildfire	1.179	0.620	1.902	0.058
log(total_deaths)	0.161	0.065	2.460	0.014
log(no_injured)	0.422	0.058	7.306	0.000

After displaying the model coefficients and p-values, we realized that the coefficient for log(no_affected) was not statistically significant at all because of its p-value of 0.853, which is much greater than 0.05. We decided to explore one last model, a model with all the previous predictors as additive effects but without log(no_affected).

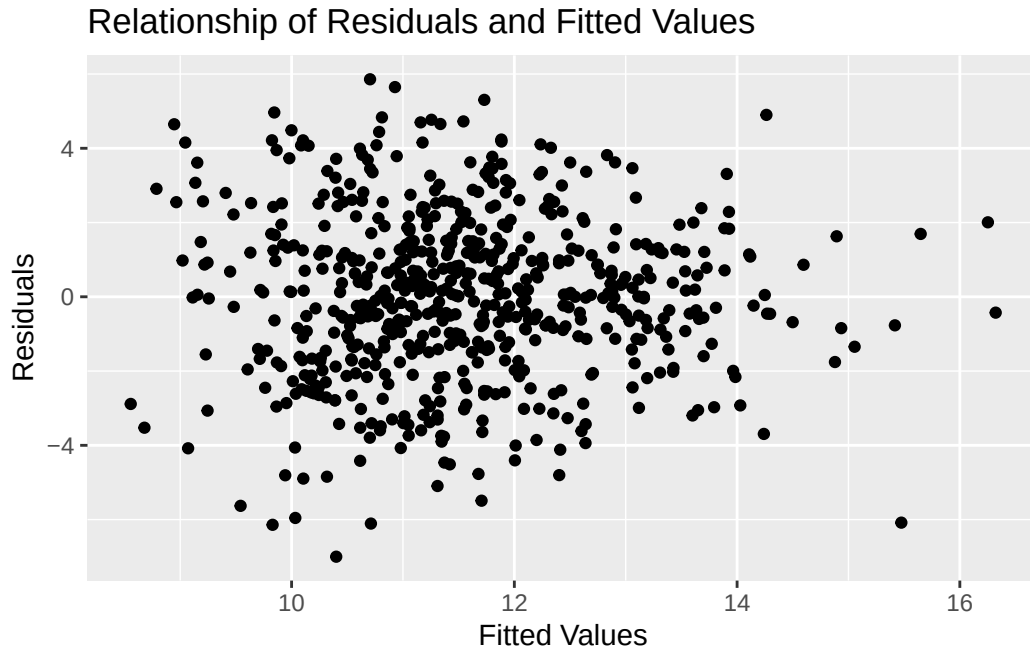
```
# A tibble: 1 x 3
  .metric .estimator .estimate
  <chr>   <chr>         <dbl>
1 rmse    standard         2.59

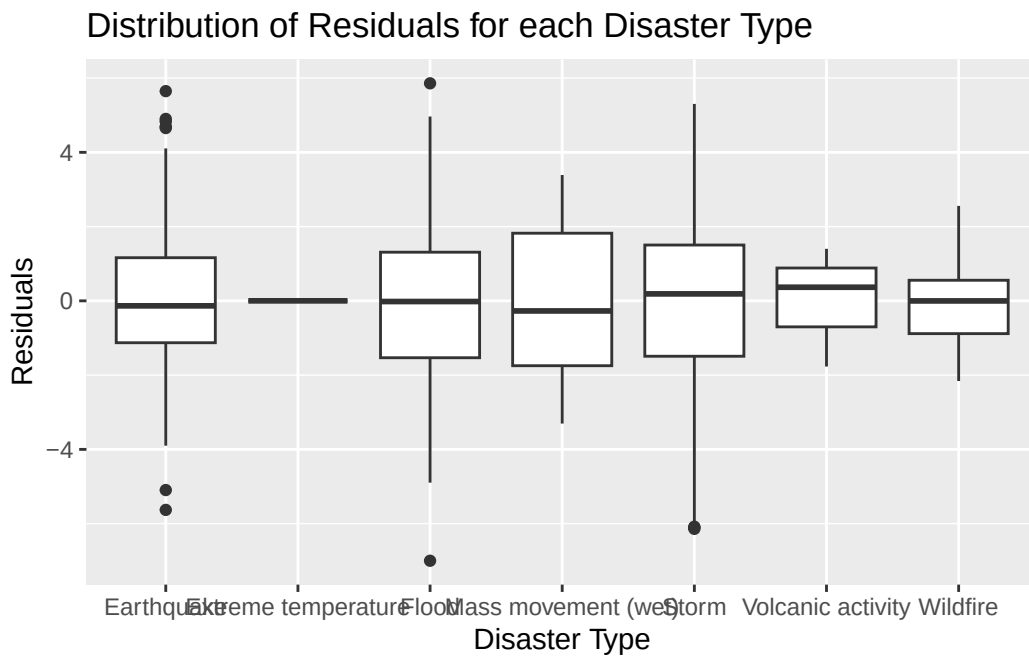
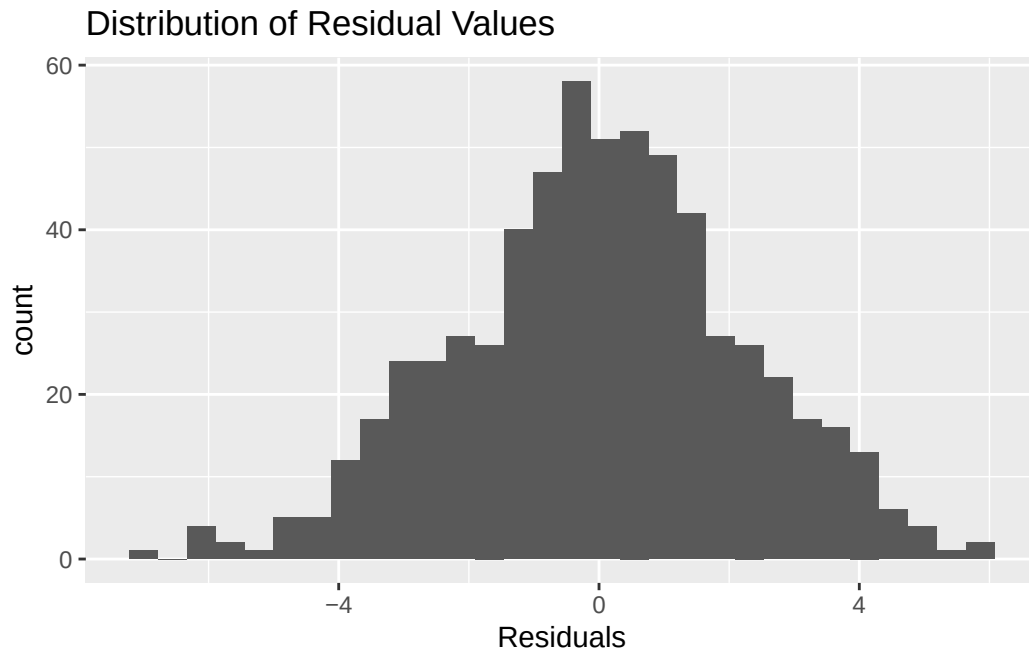
# A tibble: 1 x 3
  .metric .estimator .estimate
  <chr>   <chr>         <dbl>
1 rmse    standard         2.59
```

After evaluating the performance of the two models on the testing data, we discovered that the model without log(no_affected) performed slightly better with a lower RMSE value of 2.592011 compared to 2.593128. Although this is not significantly different, the model without the log(no_affected) has less predictors and therefore, simpler, which makes it the more optimal model.

term	estimate	std.error	statistic	p.value
(Intercept)	8.952	0.325	27.506	0.000
log_duration_days	0.287	0.106	2.718	0.007
disaster_typeExtreme temperature	-0.037	2.264	-0.016	0.987
disaster_typeFlood	-0.596	0.372	-1.601	0.110
disaster_typeMass movement (wet)	-1.103	0.668	-1.651	0.099
disaster_typeStorm	0.004	0.281	0.013	0.990
disaster_typeVolcanic activity	0.106	1.310	0.081	0.936
disaster_typeWildfire	1.161	0.612	1.897	0.058
log(total_deaths)	0.166	0.059	2.787	0.005
log(no_injured)	0.423	0.057	7.365	0.000

We then evaluated the model diagnostics for the chosen model.





```
# A tibble: 0 x 11
# i 11 variables: log_total_damage_000_us <dbl>, log_duration_days <dbl>,
#   disaster_type <chr>, log(total_deaths) <dbl>, log(no_injured) <dbl>,
```

```
# .fitted <dbl>, .resid <dbl>, .hat <dbl>, .sigma <dbl>, .cooks_d <dbl>,  
# .std.resid <dbl>
```

Results

For the chosen model, the model diagnostics were all satisfied and there were no influential points in the data set.

- The linearity condition is satisfied because the points are randomly scattered in the plot of fitted values vs. residual values. We can not guess whether the residual would be positive or negative for any predicted value.
- The constant variance condition is satisfied because the vertical spread is approximately equal for each predicted value. Although the cluster of points appear to form a fan shape, there are points outside of the fan shape pattern that allow this condition to be satisfied.
- The normality condition is satisfied because the distribution is unimodal and roughly symmetric and thus, similar to a normal distribution.
- The independence condition is satisfied because we are told that a disaster is included in the data set if at least one of the following criteria are met: 10+ people killed, 100+ people affected, declaration of a state of emergency, and solicitation of international assistance. Each disaster is an independent event as well. Furthermore, in the graph displaying the distribution of the residuals for each disaster type, we see that there is no correlation between the residuals and disaster type.

The statistically significant predictors in the model provide meaningful insight into what drives economic damage from natural disasters:

1. `no_injured`:

For each 1% increase in the number of people injured, the expected total economic damage increases by approximately 0.422%, holding all other variables constant, making it the strongest and most statistically significant predictor in the model.

2. `no_deaths`:

For each 1% increase in total fatalities, the expected total economic damage increases by approximately 0.161%, holding all other variables constant, providing further evidence that human casualties are positively associated with economic losses.

3. `duration_days`:

Disaster duration also plays a meaningful role: for each 1% increase in the duration of the disaster in days, the expected total economic damage increases by approximately 0.285%, holding all other variables constant, suggesting that longer-lasting disasters tend to cause progressively greater economic losses.

4. disaster_typeWildfire:

Wildfires are expected to multiply total economic damage by a factor of 3.193 ($e^{1.161} = 3.193$) compared to earthquakes (the baseline category), holding all other variables constant, though this result is only marginally significant ($p = 0.058$).

The coefficients from the model suggest that human impact measures, particularly the number of people injured and disaster duration, are the most important observable predictors of economic damage from natural disasters. The positive relationship between injuries and economic damage makes intuitive sense: a higher injury toll signals a more intense and widespread event, which simultaneously destroys physical infrastructure and strains public health systems — all of which translate into greater economic costs. The significance of disaster duration further supports this logic, as prolonged events allow for more cumulative destruction of assets and livelihoods.

Discussion

Include that we removed data with `drop_na()`.